

Motor Datasheet: WxF-115x26SL

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	113.0	mm
Machine Inner Diameter	75.6	mm
Machine Thickness / Width	26.4	mm
Air Gap	0.50	mm
Fill Factor (input)	0.60	-
Slot Fill Factor (calculated)	Not calculated	-
Slot Width (tooth to tooth)	3.709	mm
Available Coil Space	1.855	mm
Assumed Coil Width (simulation)	1.655	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	869.5	g
Rotor Mass	194.7	g
Stator Mass	674.8	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	4457.4	g·cm ²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	58.738	A
Supply Current (Peak)	83.069	A
Id (Peak)	-0.000	A
Iq (Peak)	83.069	A
Phase Advance Angle	0.0	°elec
Phase Resistance	9.294	mOhm
D-Axis Inductance (Ld)	35.851	μH
Q-Axis Inductance (Lq)	34.830	μH
Terminal Inductance (Ltt = Ld + Lq)	70.681	μH
PM Flux Linkage	3.946	mWb
Frequency at operating point (electrical)	791.667	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V
Power Factor	0.806	-

Performance

Parameter	Value	Unit
Torque (Continuous)	9.99	Nm
Speed	2500.0	rpm
Dimensionless motor constant	13.9	-
BackEMF Constant (Ke, phase peak)	7.9	mV/rpm
Motor Constant (Km, only resistive losses)	1.02	Nm/√W
Specific Motor Constant (Km / mass)	1.17	Nm/√W/kg
Torque Constant (Kt, peak, no-load)	114.6	mNm/A

Torque Ripple	4.8	%
Saturation	0.0	%

Losses

Parameter	Value	Unit
Total Losses	185.0	W
Resistive Losses	96.2	W
Iron Core Losses	63.8	W
Magnet Eddy Current Losses	9.9	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

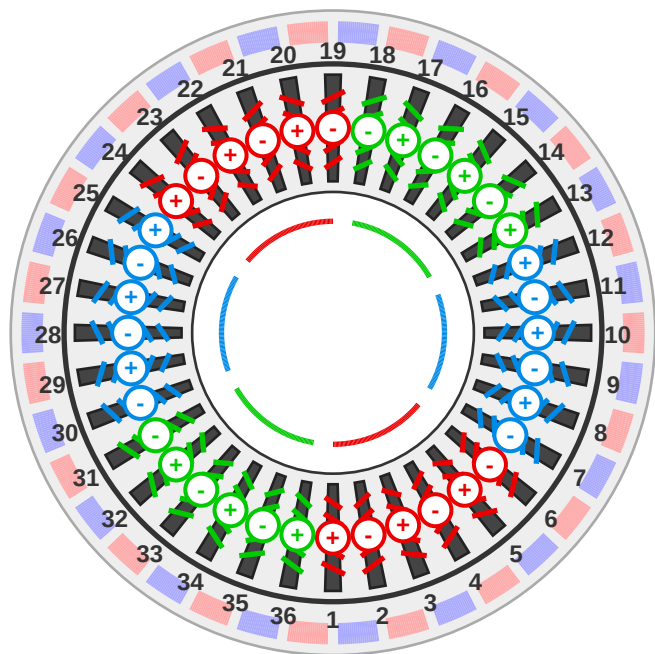
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	36	-
Pole Pairs	19	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	4.0	-
Number of Phase Turns (all phases in series)	48.0	-

Winding Configuration

Schema: **AaAaAabBbBbBcCcCcCcaAaAaABbBbBbcCcCcC**

Concentrated Winding | Balanced | Cogging: 684/rev | Ripple: 684/rev



Materials

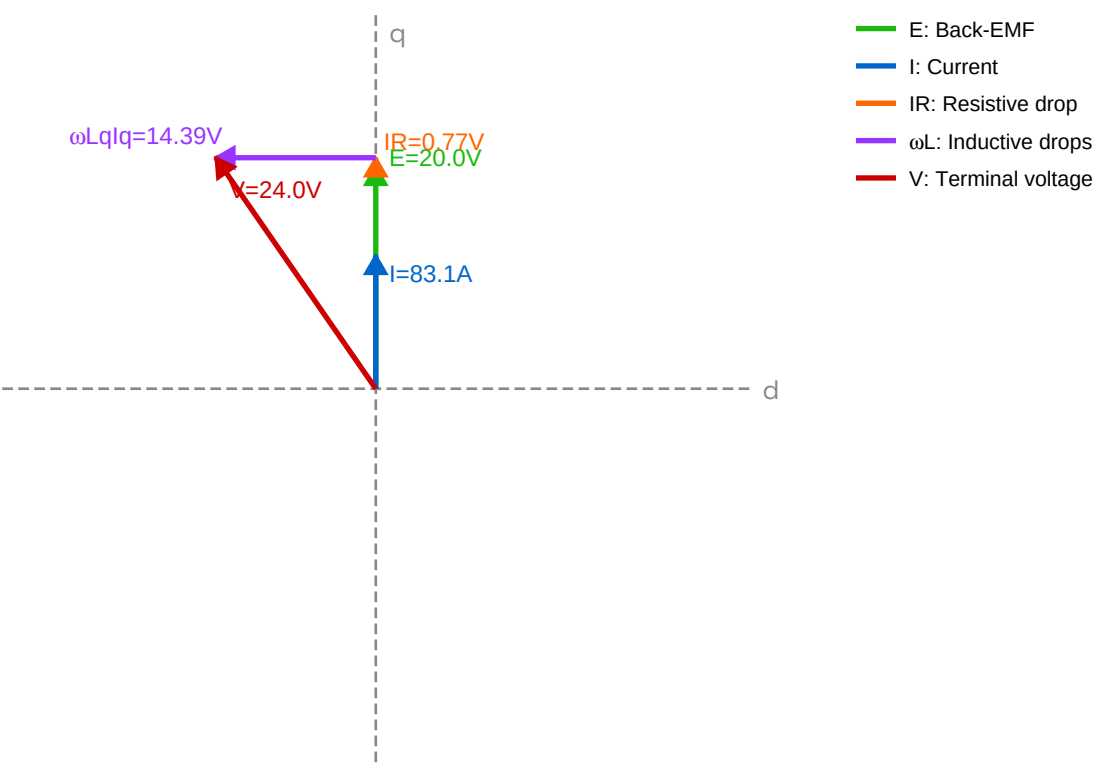
Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-
Rotor Backiron	AISI 1010 Annealed	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	20.008	V
Resistive Drop (Phase, Peak)	0.772	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	14.392	V

Inductive Drop Total (Peak)	14.392	V
Terminal Voltage (Phase, Peak)	24.000	V
Back-EMF Voltage (Line-to-Line, Peak)	34.655	V
Resistive Drop (Line-to-Line, Peak)	1.337	V
Inductive Drop (Line-to-Line, Peak)	24.927	V
Terminal Voltage (Line-to-Line, Peak)	41.569	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	481.99	N
Magnet Axial Force Positive Maximum	0.00	N
Magnet Axial Force Positive Minimum	0.00	N
Magnet Axial Force Negative Maximum	-73.54	N
Magnet Axial Force Negative Minimum	-44.82	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.